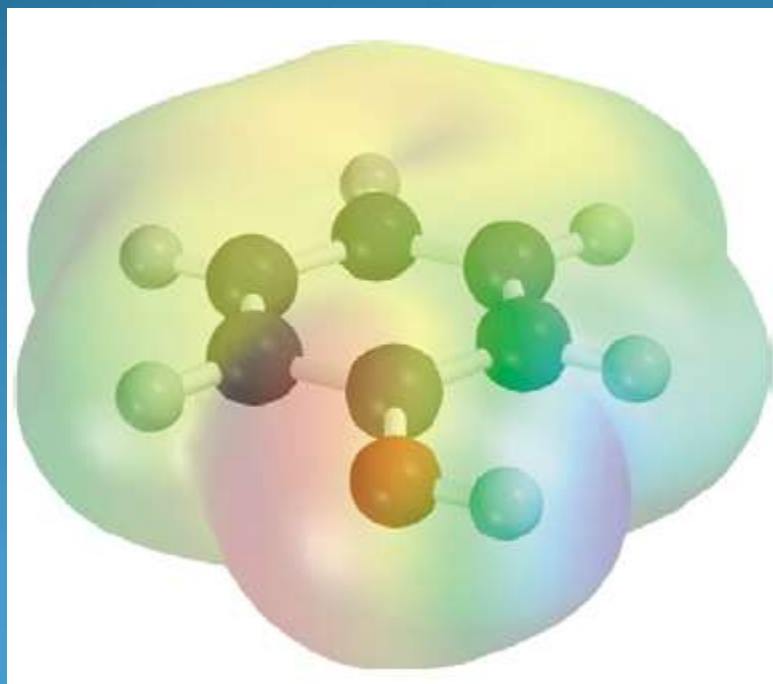
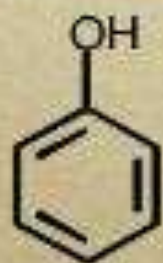




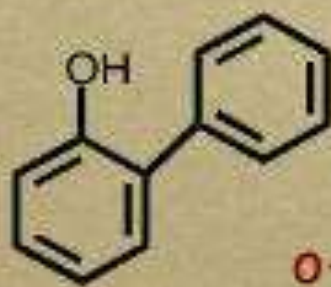
Phenols

Phenols

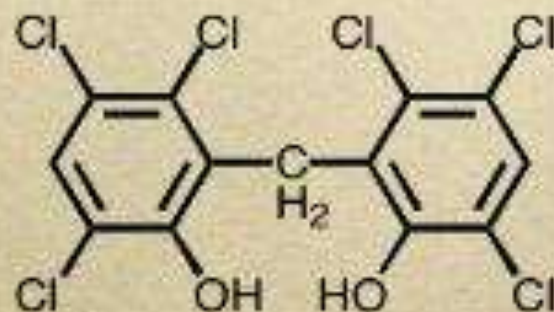
"Alcohol-like" compounds that have the hydroxyl group bound to a benzene ring



phenol



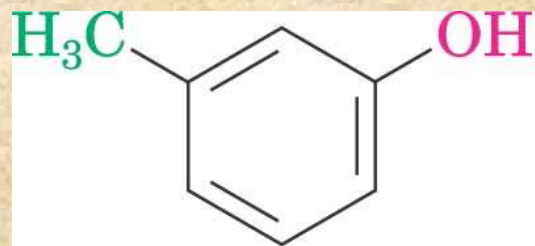
o-phenylphenol



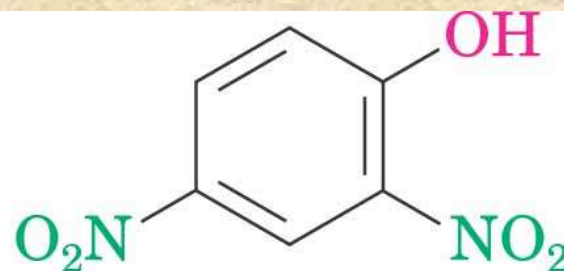
hexachlorophene

Phenols

- Use “phene” (the French name for benzene) as the parent hydrocarbon name, not benzene
- Name substituents on aromatic ring by their position from OH

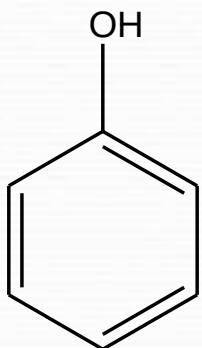


***m*-Methylphenol**
(*m*-Cresol)

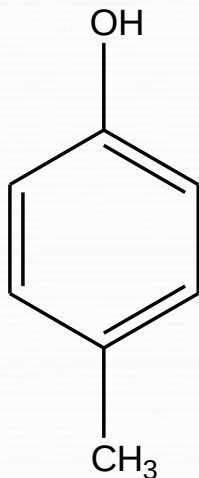


2,4-Dinitrophenol

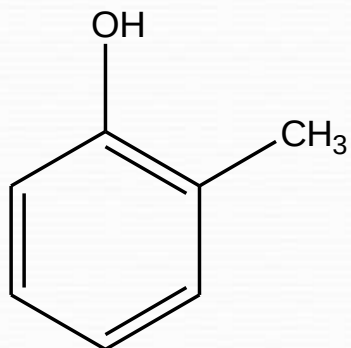
PHENOLS



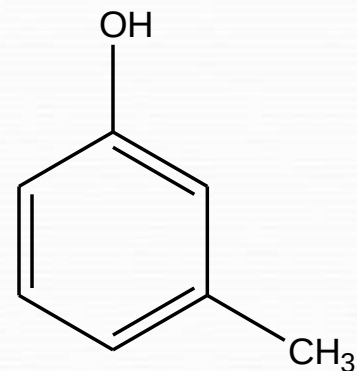
Phenol



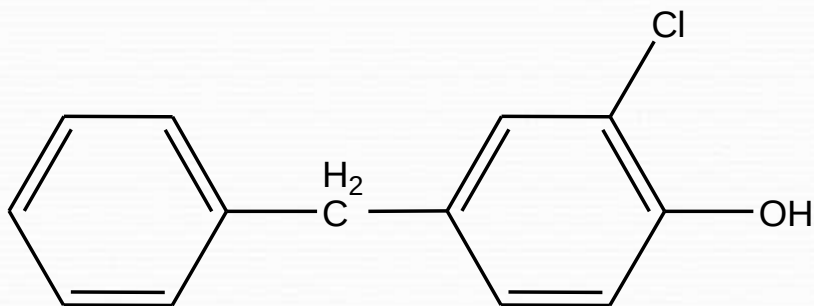
p-Cresol



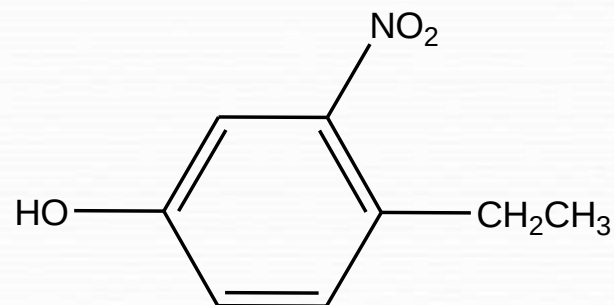
o-Cresol



m-Cresol



4-Benzyl-2-chlorophenol

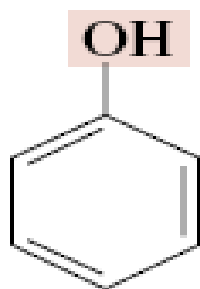


4-Ethyl-3-nitrophenol

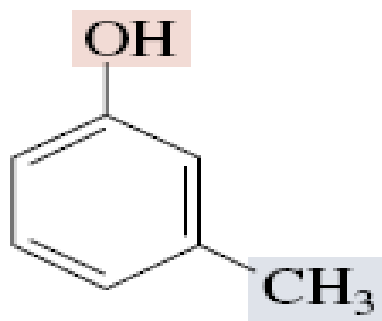
1-Nomenclature of Phenols

- Phenols are compounds that have hydroxyl group bonded directly to benzene ring.

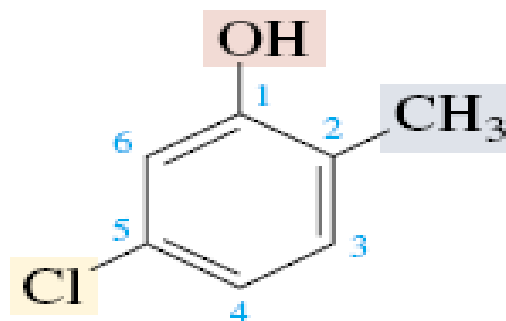
Named on basis of phenol as parent, substituents listed in alphabetical order.



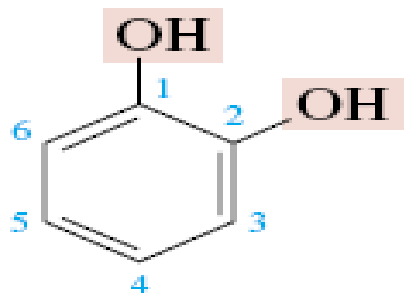
Phenol



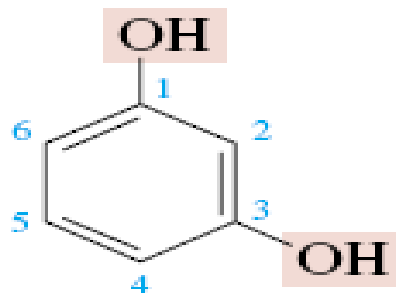
m-Cresol



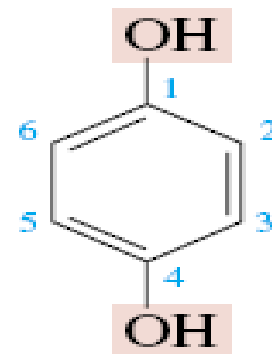
5-Chloro-2-methylphenol



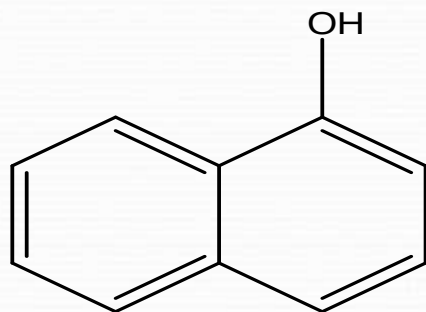
1,2-Benzenediol
(pyrocatechol)



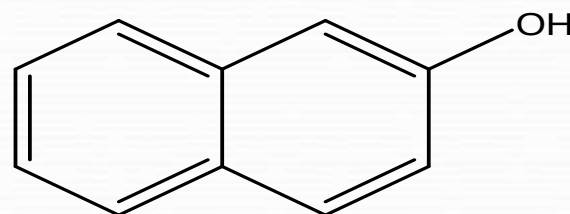
1,3-Benzenediol
(resorcinol)



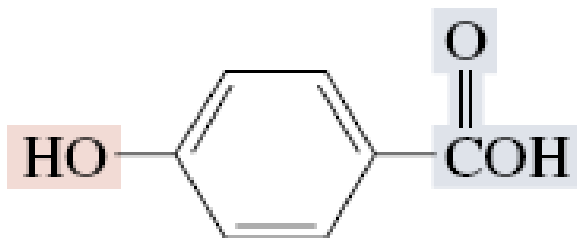
1,4-Benzenediol
(hydroquinone)



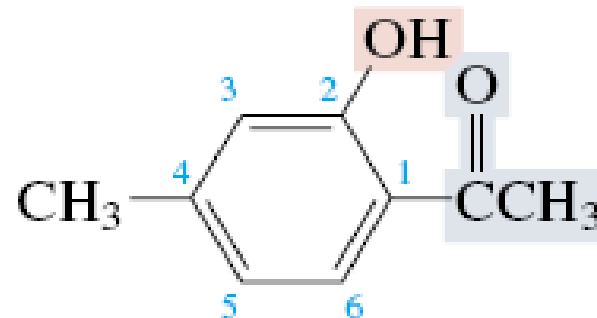
1- Naphthol



2- Naphthol



p-Hydroxybenzoic acid



2-Hydroxy-4-methylacetophenone

PROBLEM 24.1 Write structural formulas for each of the following compounds:

- | | |
|-------------------------------------|------------------------|
| (a) Pyrogallol (1,2,3-benzenetriol) | (c) 3-Nitro-1-naphthol |
| (b) <i>o</i> -Benzylphenol | (d) 4-Chlororesorcinol |

2-Physical Properties

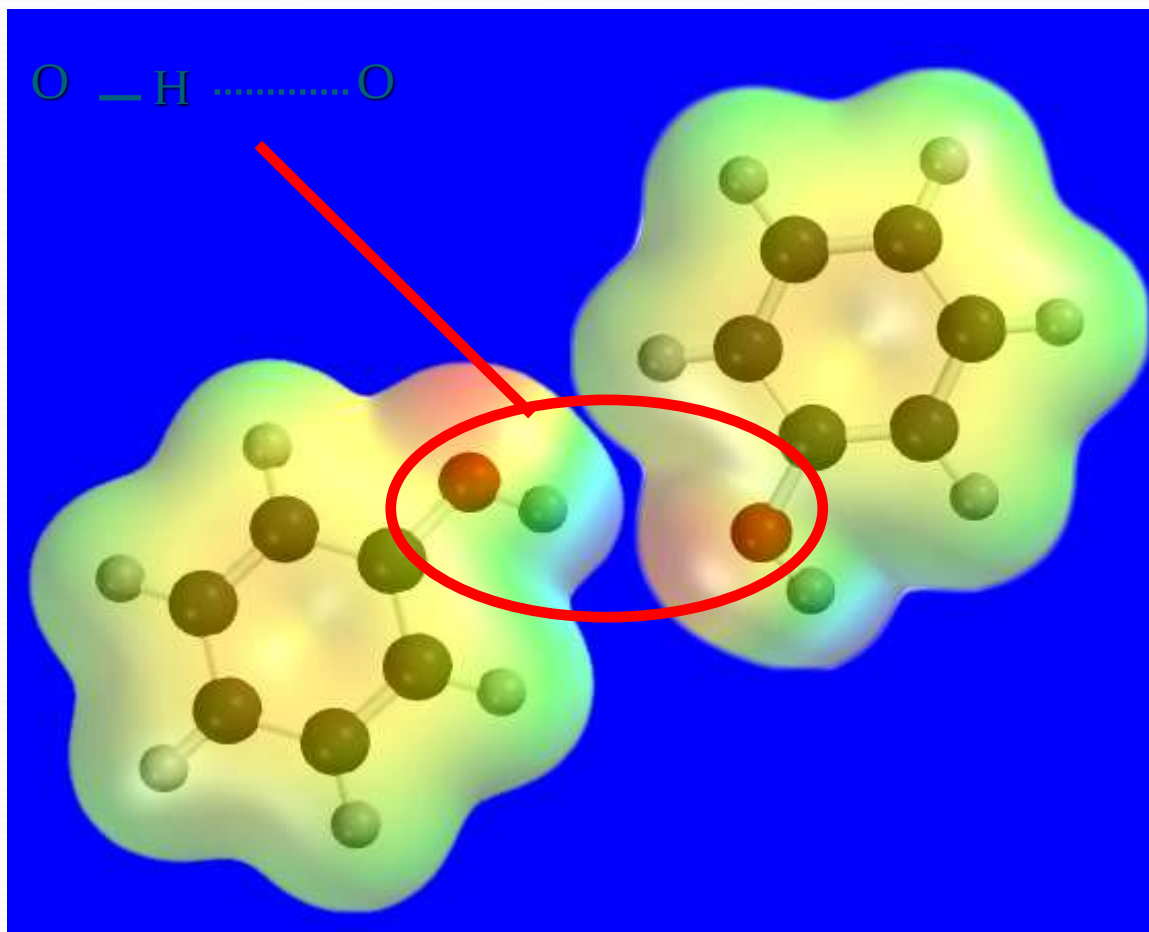
1-The OH group of phenols allows hydrogen bonding to other phenol molecules and to water.

2-Compared to compounds of similar size and molecular weight, hydrogen bonding in phenol raises its melting point, boiling point, and solubility in water.

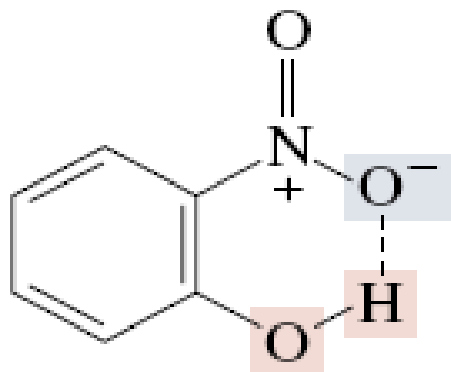
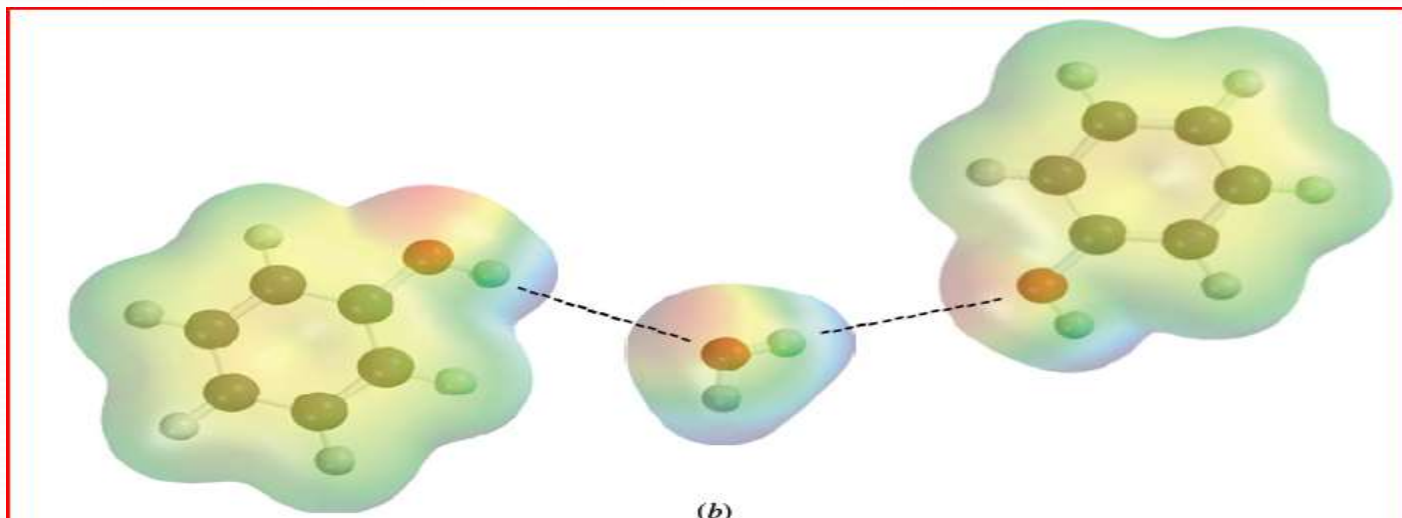
TABLE 24.1 Comparison of Physical Properties of an Arene, a Phenol, and an Aryl Halide

Physical property	Compound		
	Toluene, $\text{C}_6\text{H}_5\text{CH}_3$	Phenol, $\text{C}_6\text{H}_5\text{OH}$	Fluorobenzene, $\text{C}_6\text{H}_5\text{F}$
Molecular weight	92	94	96
Melting point	-95°C	43°C	-41°C
Boiling point (1 atm)	111°C	132°C	85°C
Solubility in water (25°C)	0.05 g/100 mL	8.2 g/100 mL	0.2 g/100 mL

Hydrogen Bonding of Phenols



Hydrogen Bonding with Water

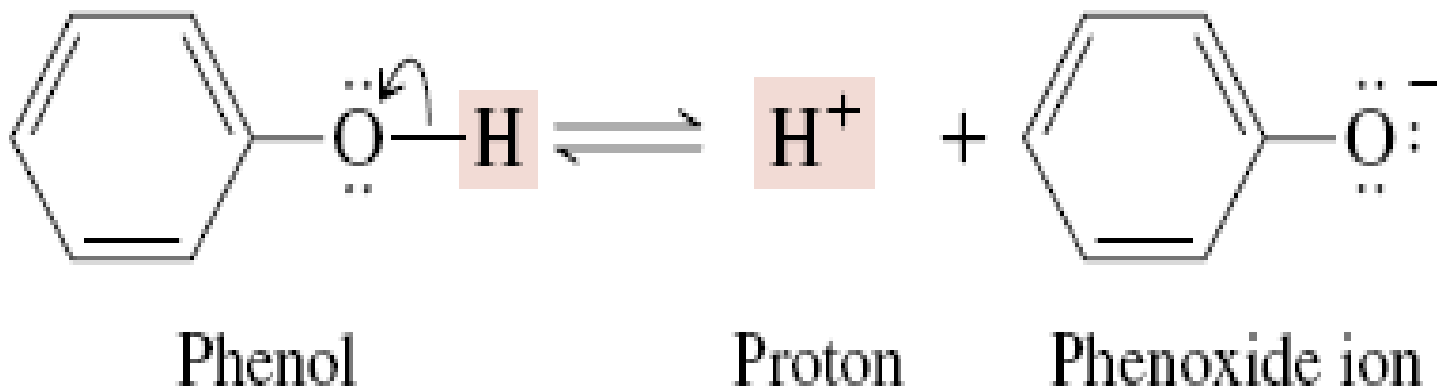


Intramolecular hydrogen bond
in *o*-nitrophenol

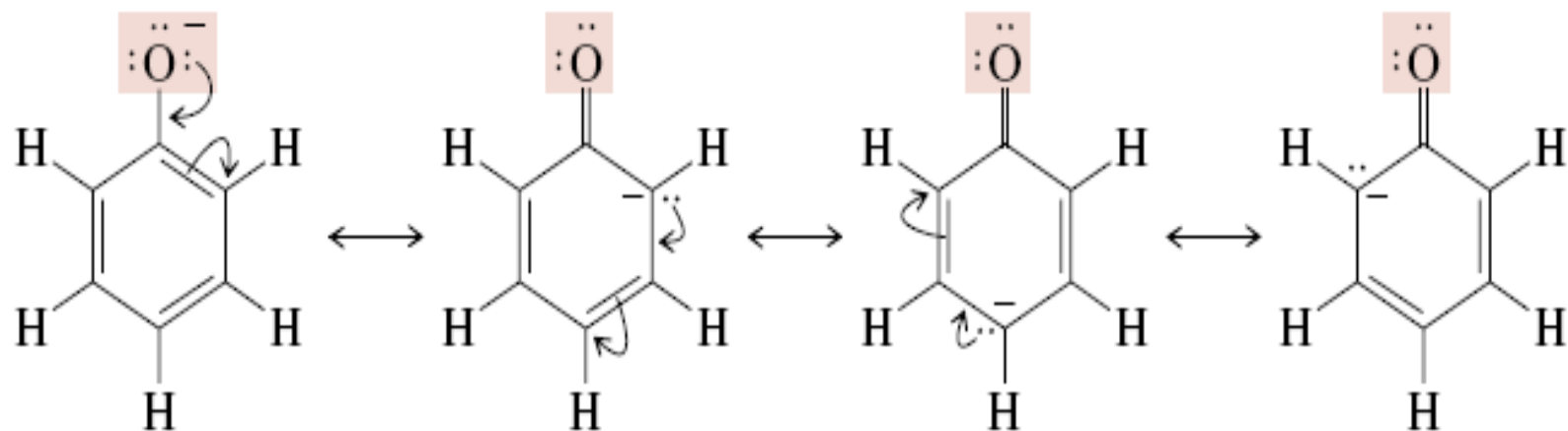
3- Acidity of Phenols

- Most characteristic property of phenols is their acidity.
- Phenols($\text{pK}_a = 10$) are more acidic than alcohols($\text{pK}_a = 16-20$) but less acidic than Carboxylic acids ($\text{pK}_a = 5$).

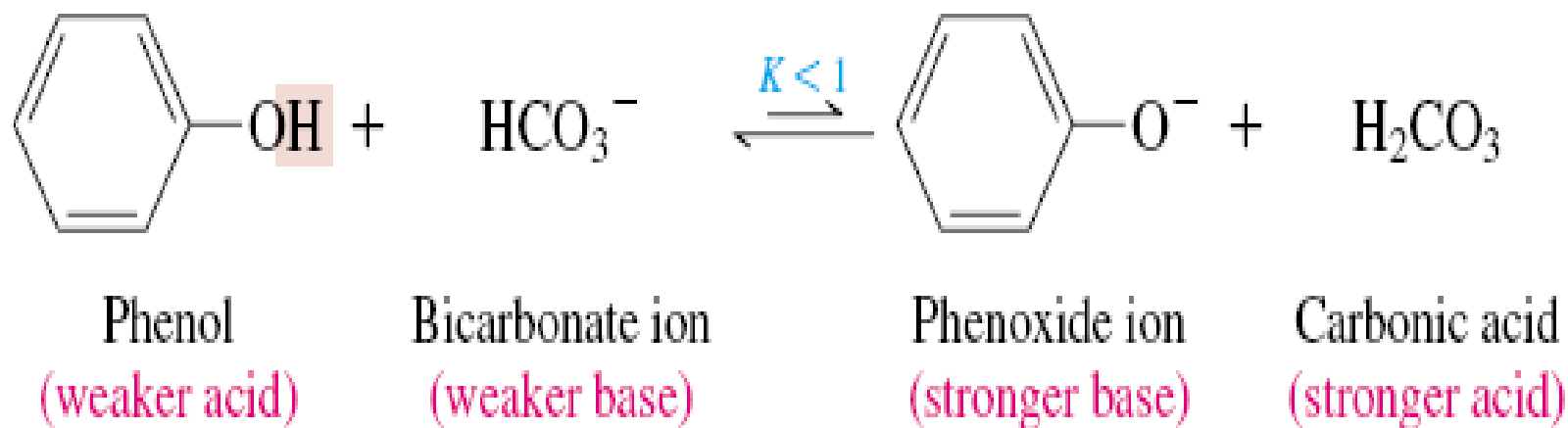
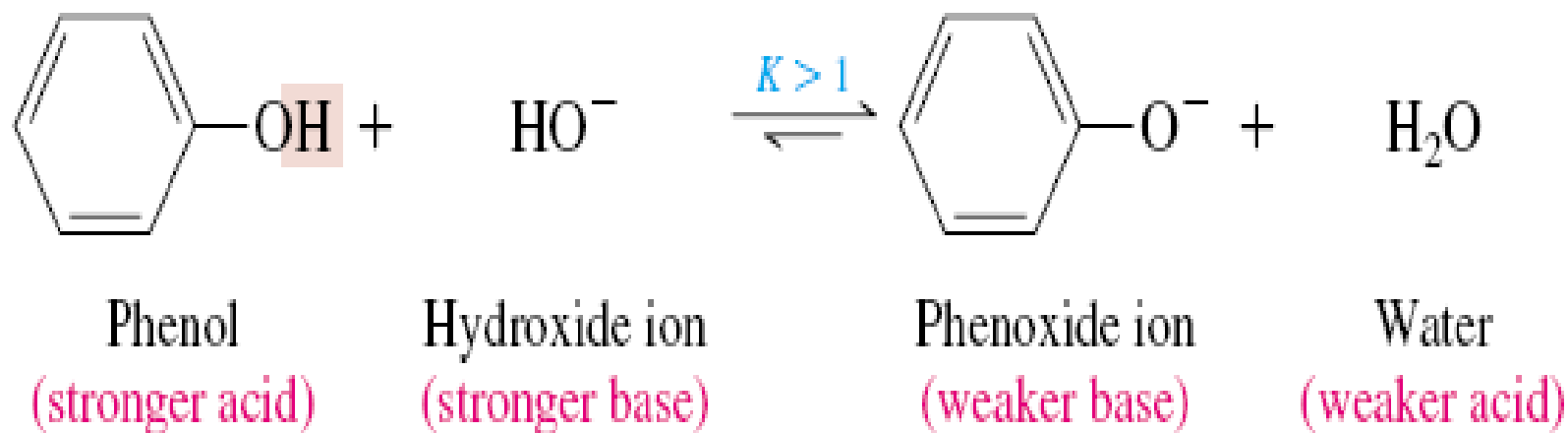
To explain why, let's compare the ionization of Ethanol and Phenols



Electron delocalization in phenoxide is represented by resonance among the structures:



The negative charge in phenoxide ion is shared by the oxygen and the carbons that are ortho and para to it. Delocalization of its negative charge strongly stabilizes phenoxide ion.



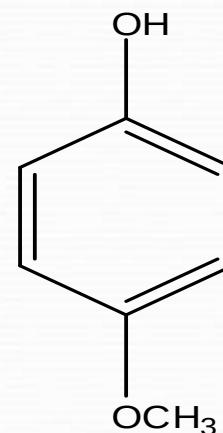
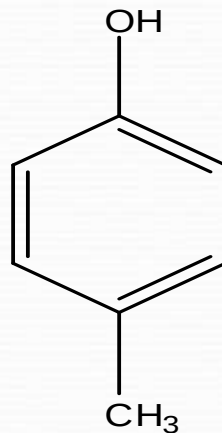
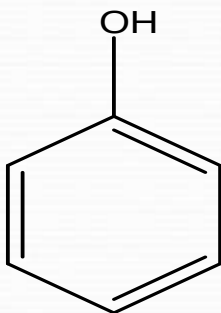
Substituent Effects on the Acidity of Phenols.

1-Electron-releasing groups (CH_3 , OCH_3) have little or no effect.

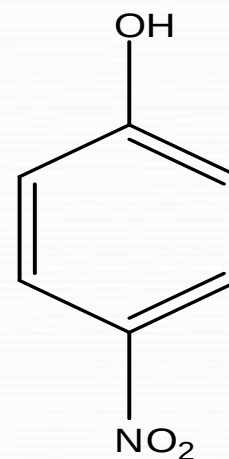
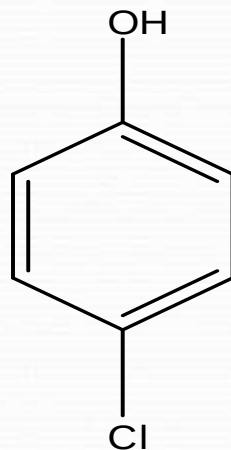
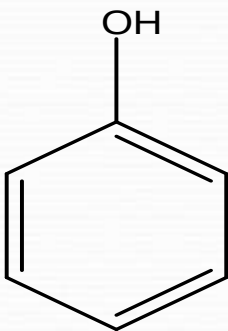
2-Electron-withdrawing groups (NO_2 , Cl , OH) increase acidity.

3- Multiple substituent of electron- withdrawing group greatly increase acidity of phenols.

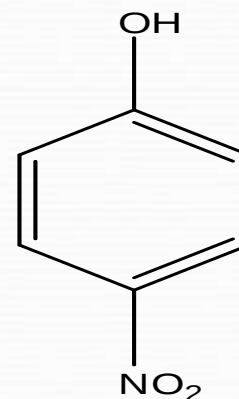
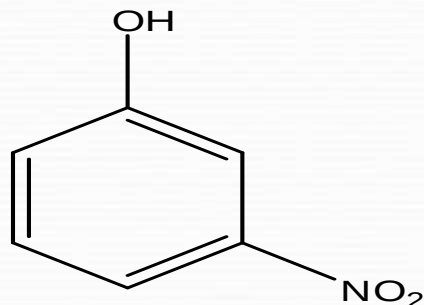
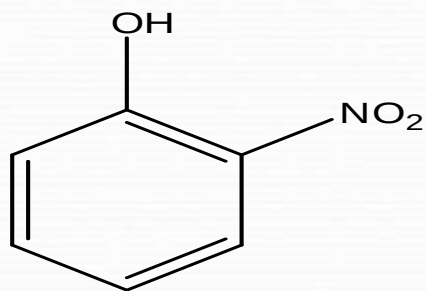
Electron-releasing groups have little or no effect.



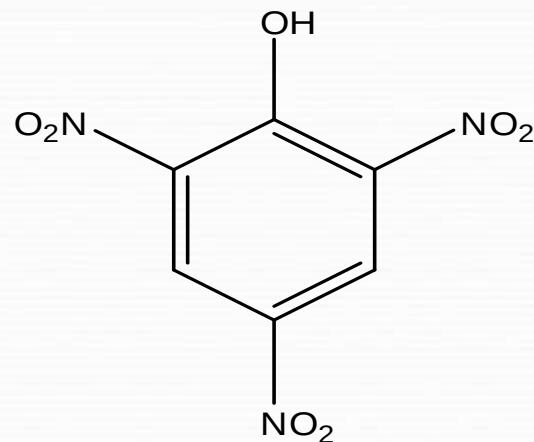
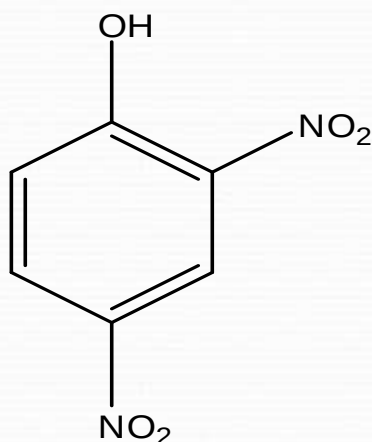
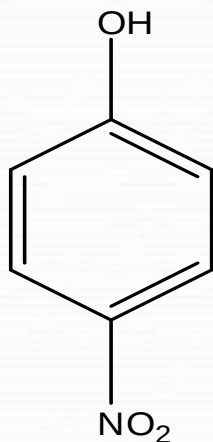
Electron-withdrawing groups increase acidity.



Effect of electron-withdrawing groups is most pronounced at ortho and para positions.



Effect of strong electron-withdrawing groups is cumulative.



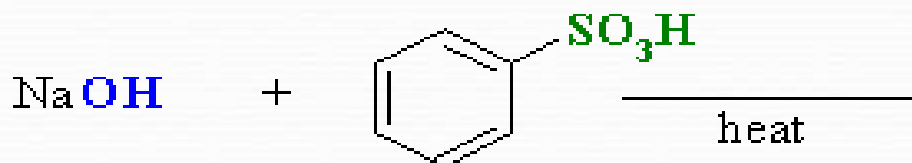
Compound name	Ionization constant K_a	pK_a
Monosubstituted phenols		
Phenol	1.0×10^{-10}	10.0
<i>o</i> -Cresol	4.7×10^{-11}	10.3
<i>m</i> -Cresol	8.0×10^{-11}	10.1
<i>p</i> -Cresol	5.2×10^{-11}	10.3
<i>o</i> -Chlorophenol	2.7×10^{-9}	8.6
<i>m</i> -Chlorophenol	7.6×10^{-9}	9.1
<i>p</i> -Chlorophenol	3.9×10^{-9}	9.4
<i>o</i> -Methoxyphenol	1.0×10^{-10}	10.0
<i>m</i> -Methoxyphenol	2.2×10^{-10}	9.6
<i>p</i> -Methoxyphenol	6.3×10^{-11}	10.2
<i>o</i> -Nitrophenol	5.9×10^{-8}	7.2
<i>m</i> -Nitrophenol	4.4×10^{-9}	8.4
<i>p</i> -Nitrophenol	6.9×10^{-8}	7.2
Di- and trinitrophenols		
2,4-Dinitrophenol	1.1×10^{-4}	4.0
3,5-Dinitrophenol	2.0×10^{-7}	6.7
2,4,6-Trinitrophenol	4.2×10^{-1}	0.4

Sources of Phenols

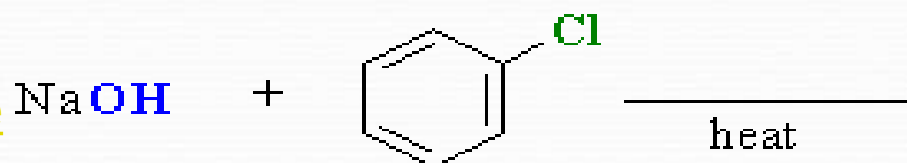
- Phenols was first isolated from coal tar.
- Phenol is an important industrial chemical.
- Major use is in phenolic resins for adhesives and plastics.

preparation of Phenols

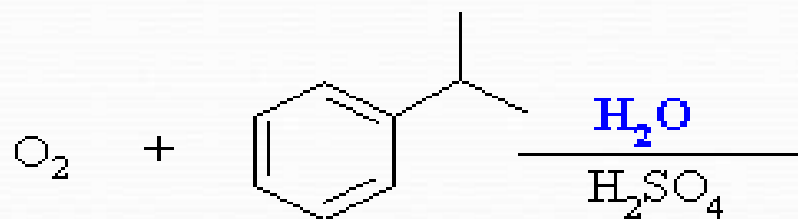
Reaction of benzene sulfonic acid with base



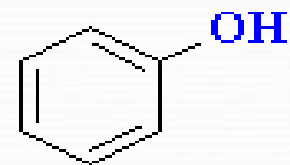
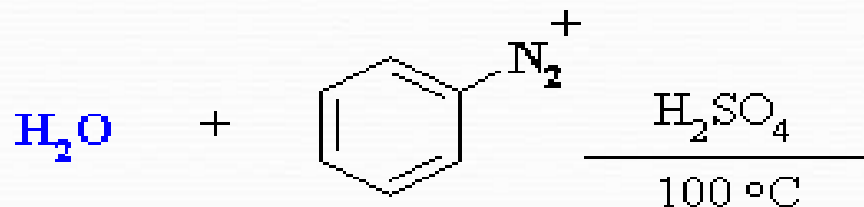
Reaction of chlorobenzene with base



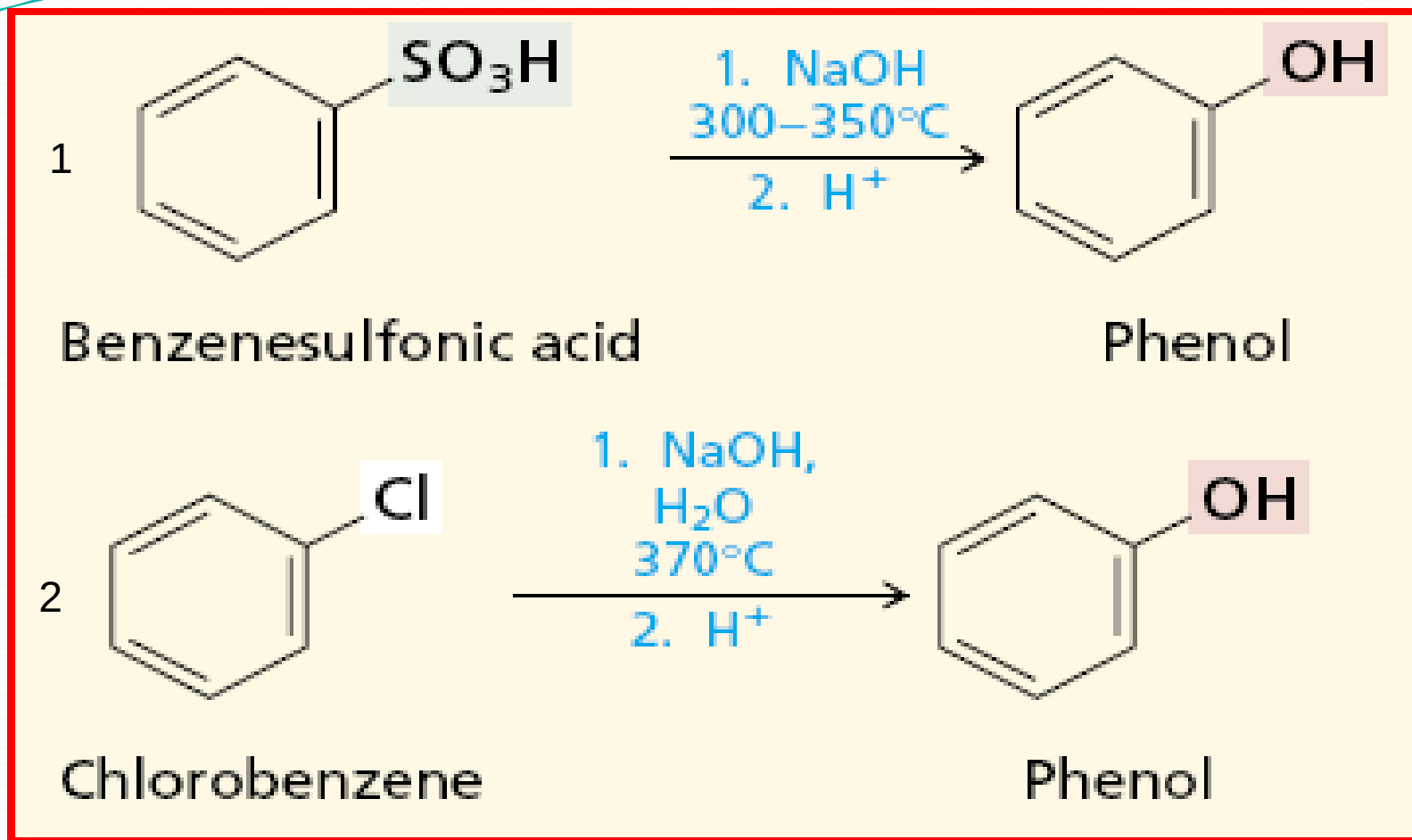
Acidic oxidation of cumene



Hydrolysis of diazonium salts



Industrial preparation of Phenols

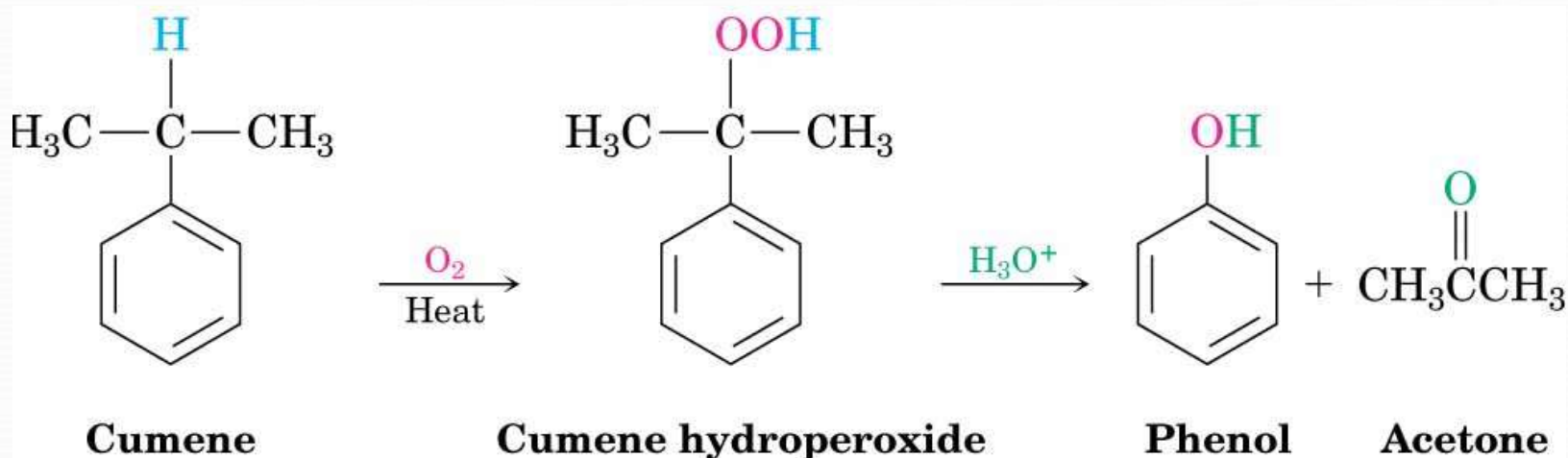


1. This represents the oldest method for the industrial preparation of phenol.
 - The reaction occurs via the addition-elimination mechanism with SO_3^{2-} functioning as the leaving group.
2. The reaction occurs via the elimination-addition mechanism via a benzyne intermediate

Industrial preparation of Phenols

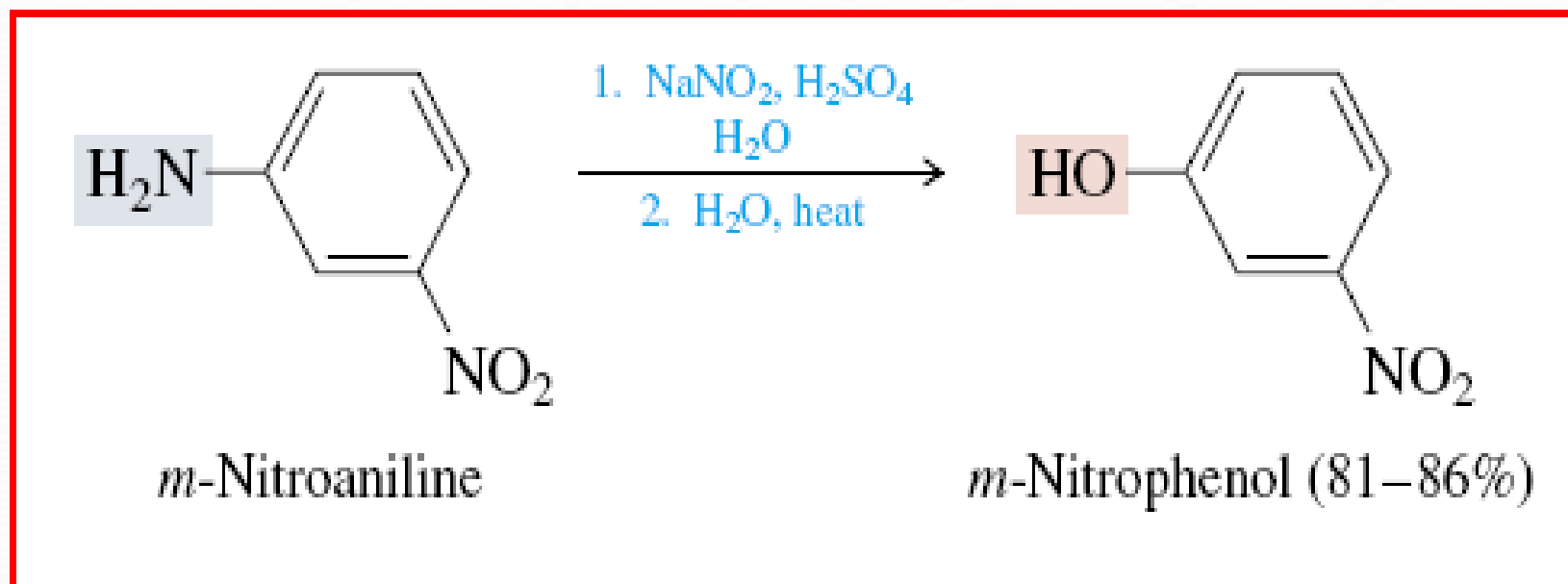
Industrial process from readily available cumene

Forms cumene hydroperoxide with oxygen at high temperature



Laboratory Synthesis of Phenols

- The most important synthesis of phenol in the laboratory is from amine by hydrolysis of their corresponding diazonium salt



Aryl diazonium salts are prepared by reaction of aryl amines with nitrous acid, HNO_2

PHENOL - REACTIONS OF THE OH GROUP

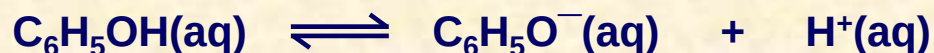
Water

phenol is a weak acid

it dissolves very slightly in water to form a weak acidic solution

it is a stronger acid than aliphatic alcohols

the ring helps weaken the O-H bond and stabilises the resulting anion



NaOH

phenol reacts with sodium hydroxide to form a salt - sodium phenoxide

it is ionic and water soluble

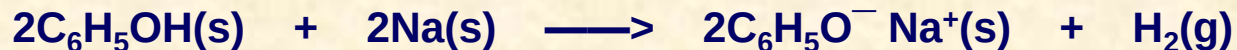


Sodium

phenol reacts with sodium to form an ionic salt - sodium phenoxide

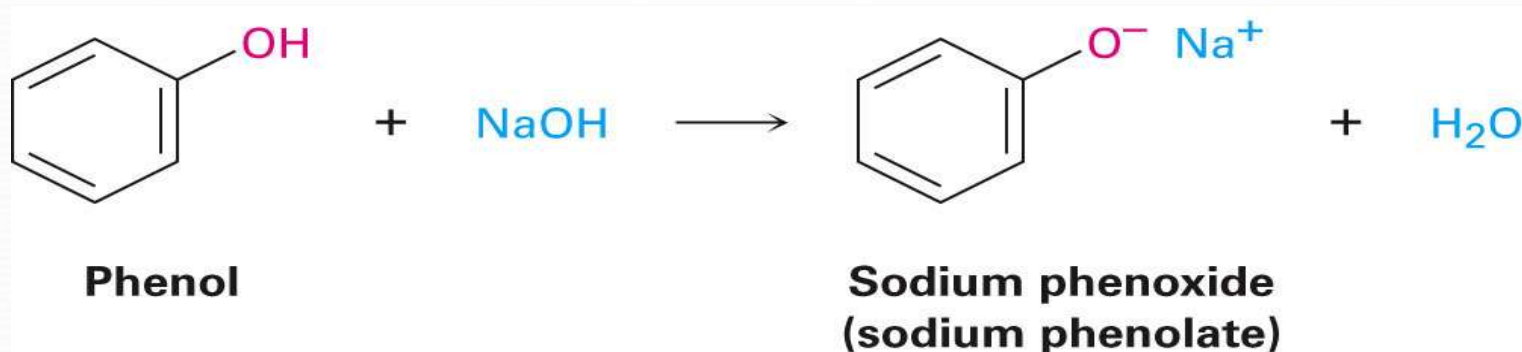
hydrogen is also produced

this reaction is similar to that with aliphatic alcohols such as ethanol



Phenol Acidity

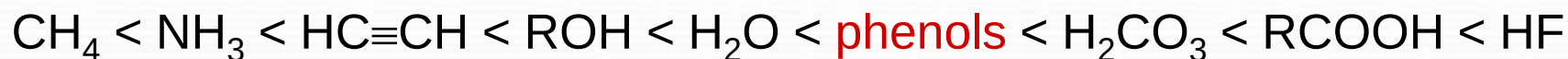
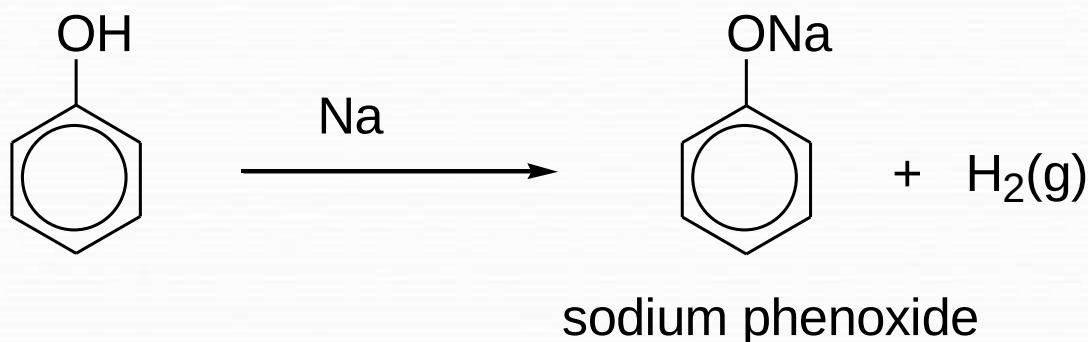
- Phenols ($\text{pK}_a \sim 10$) are much more acidic than alcohols ($\text{pK}_a \sim 16$) due to resonance stabilization of the phenoxide ion
- Phenols react with NaOH solutions (but alcohols do not), forming salts that are soluble in dilute aqueous solution



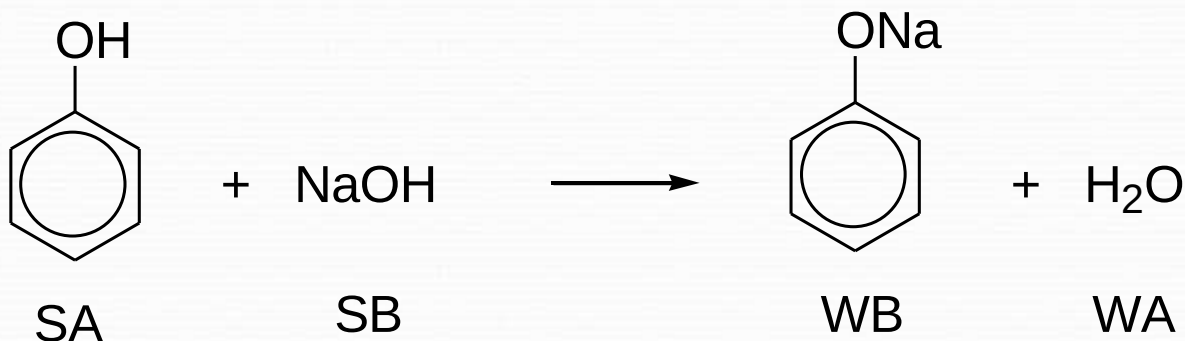
Phenol Acidity

AS ACIDS:

with active metals:

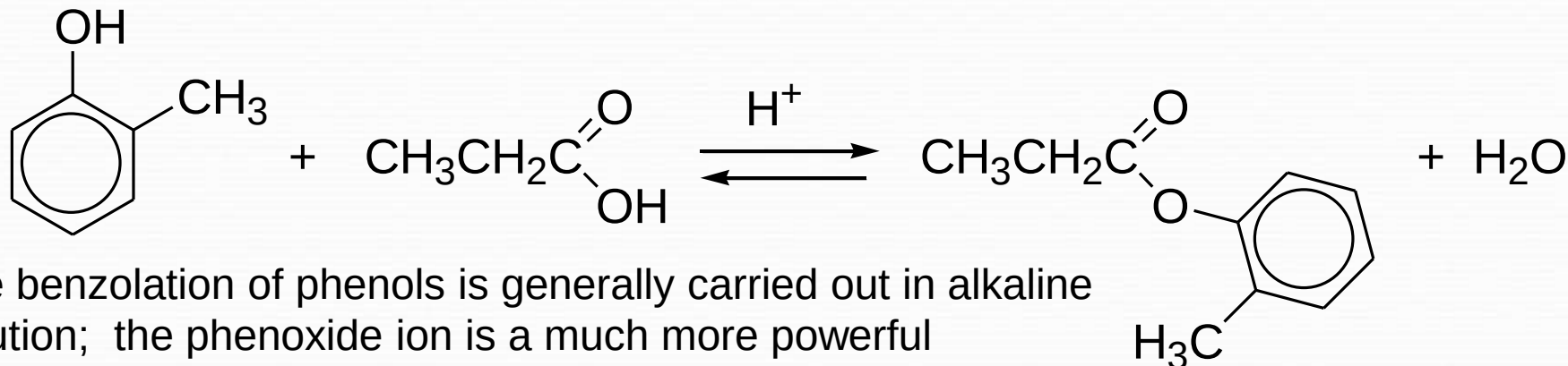


with bases:

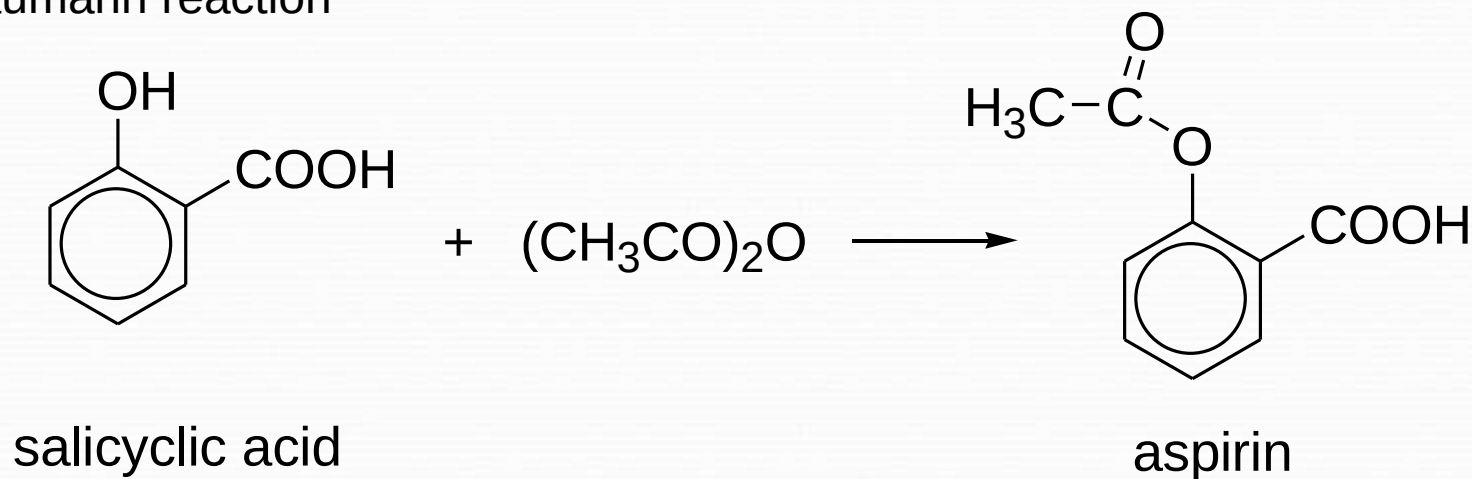


Phenol Acidity

2. ester formation (similar to alcohols)



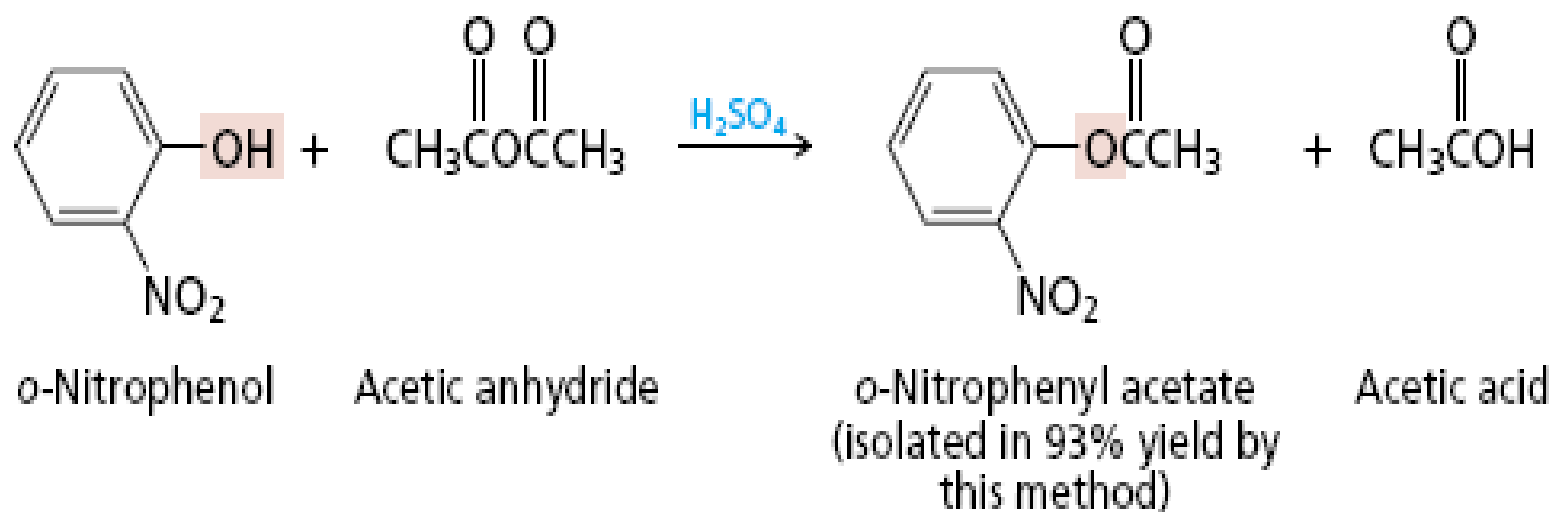
The benzoylation of phenols is generally carried out in alkaline solution; the phenoxide ion is a much more powerful nucleophilic reagent than phenol itself. This is known as Schotten-Baumann reaction



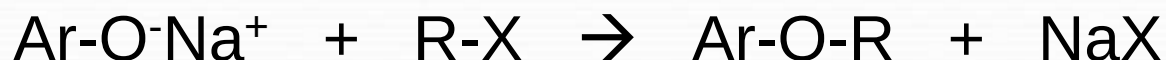
PROBLEM 24.7 Write chemical equations expressing each of the following:

- (a) Preparation of *o*-nitrophenyl acetate by sulfuric acid catalysis of the reaction between a phenol and a carboxylic acid anhydride.
- (b) Esterification of 2-naphthol with acetic anhydride in aqueous sodium hydroxide
- (c) Reaction of phenol with benzoyl chloride

SAMPLE SOLUTION (a) The problem specifies that an acid anhydride be used; therefore, use acetic anhydride to prepare the acetate ester of *o*-nitrophenol:

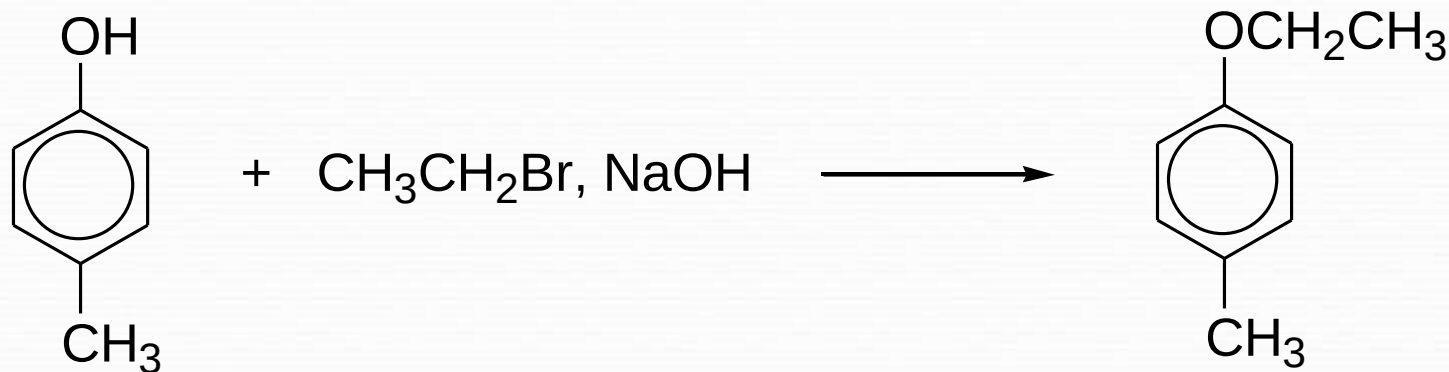


3. Ether formation (Williamson Synthesis)



note: R-X must be 1° or CH₃

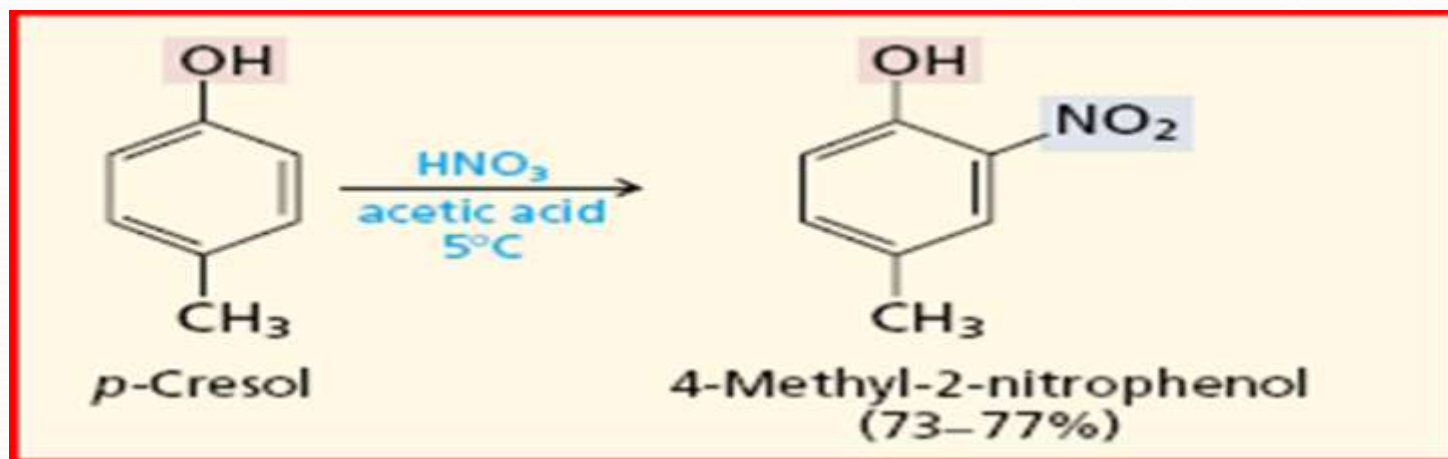
Because phenols are more acidic than water, it is possible to generate the phenoxide *in situ* using NaOH.



5- Reactions of Phenols (C-Ring)

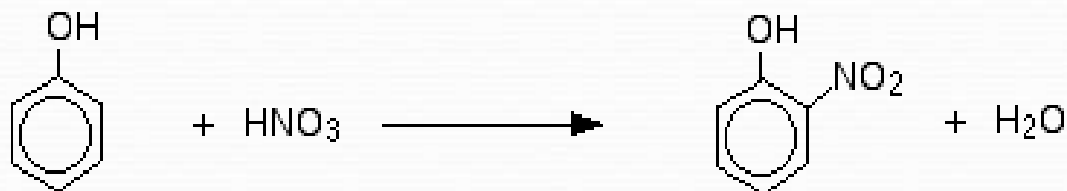
A hydroxyl group is very powerful activating substituent, and electrophilic aromatic substitution in phenol occurs far faster, and under milder condition, than in benzene.

a- Nitration

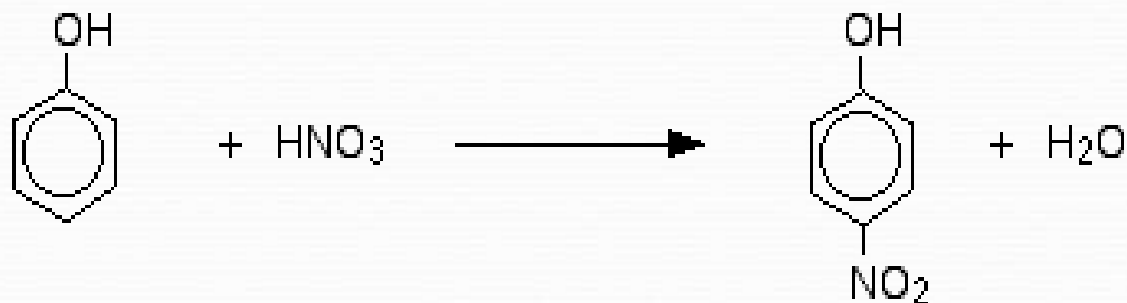


a- Nitration

Phenol reacts with dilute nitric acid at room temperature to give a mixture of 2-nitrophenol and 4-nitrophenol

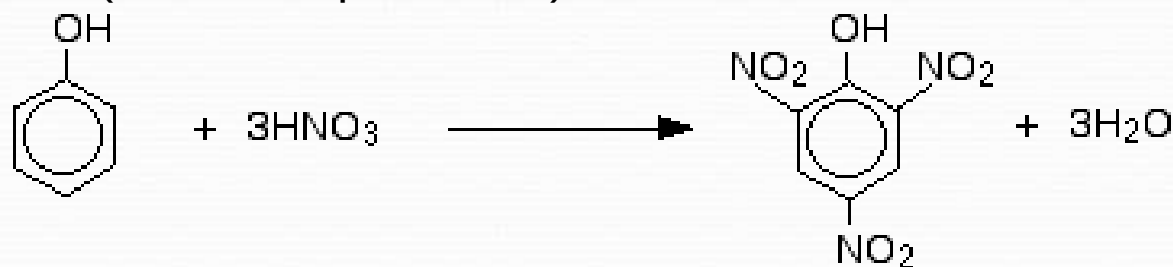


2-nitrophenol



4-nitrophenol

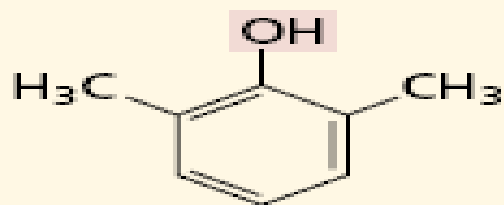
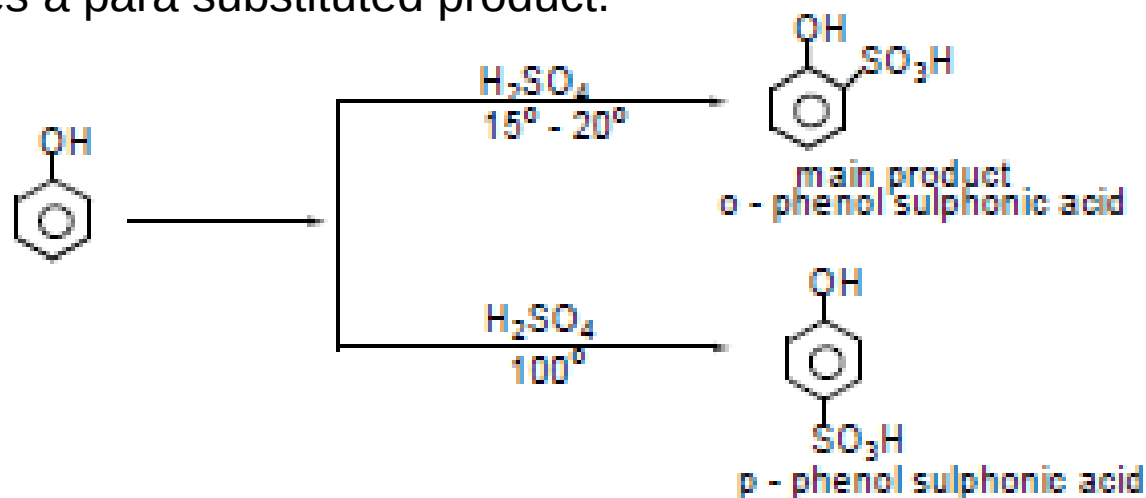
With concentrated nitric acid, more nitro groups substitute around the ring to give 2,4,6-trinitrophenol (old name: picric acid).



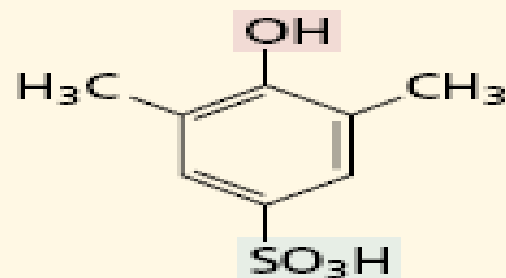
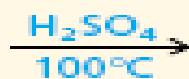
2,4,6-trinitrophenol

b- Sulfonation

This can be achieved using sulphuric acid at low or high temperature. The low temperature reaction gives ortho substituted product, while at 100°C it gives a para substituted product.



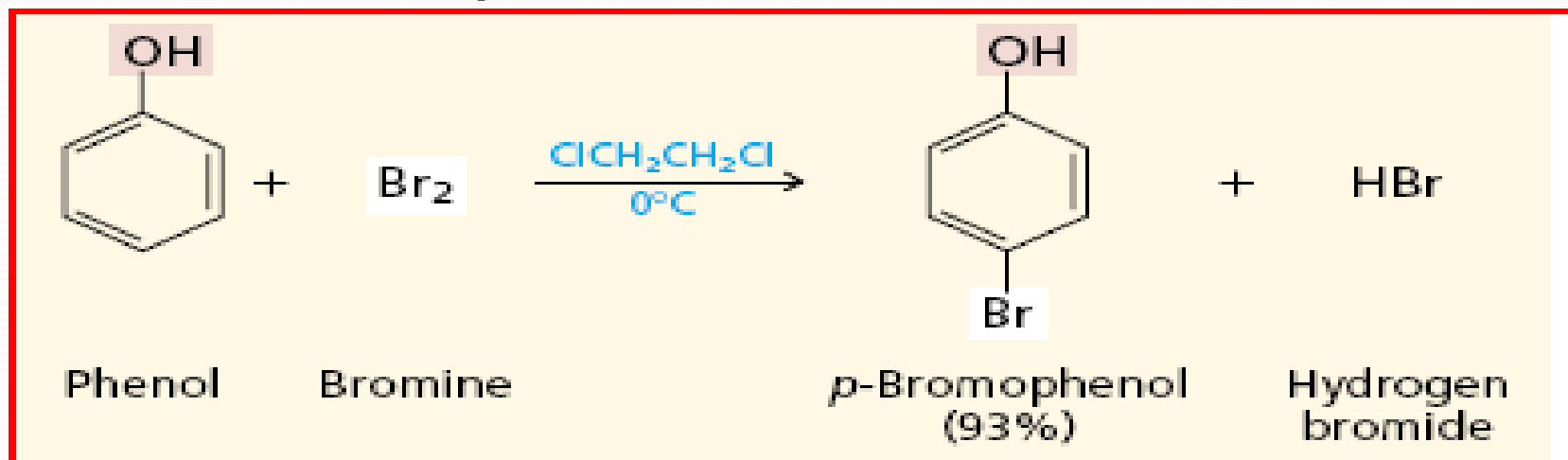
2,6-Dimethylphenol



4-Hydroxy-3,5-dimethylbenzenesulfonic acid (69%)

c- Halogenation

Bromination of phenol using bromine in carbon disulphide or 1,2-dichloroethane (non-polar solvent) will give o and p bromophenol but with water a polar solvent 2,4,6-tribromophenol is obtained.

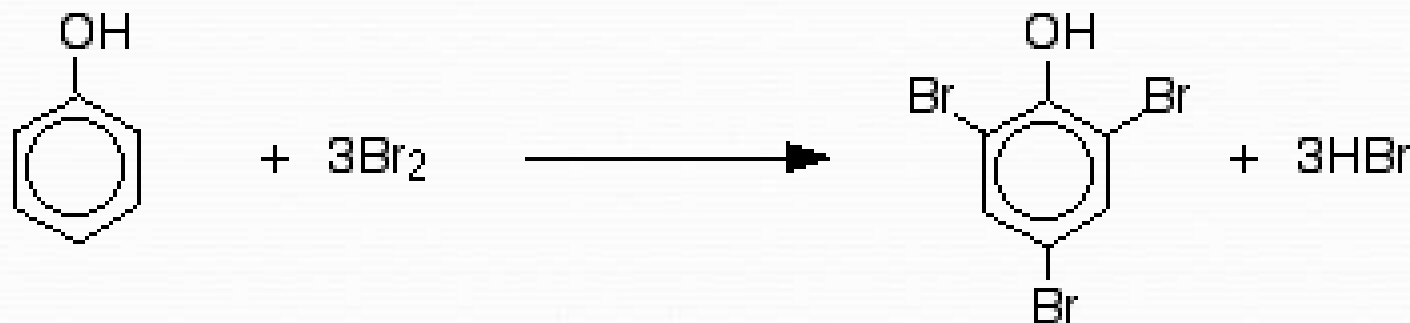


c- Halogenation

- **Reaction with bromine water**

If bromine water is added to a solution of phenol in water, the bromine water is decolourised and a white precipitate is formed which smells of antiseptic.

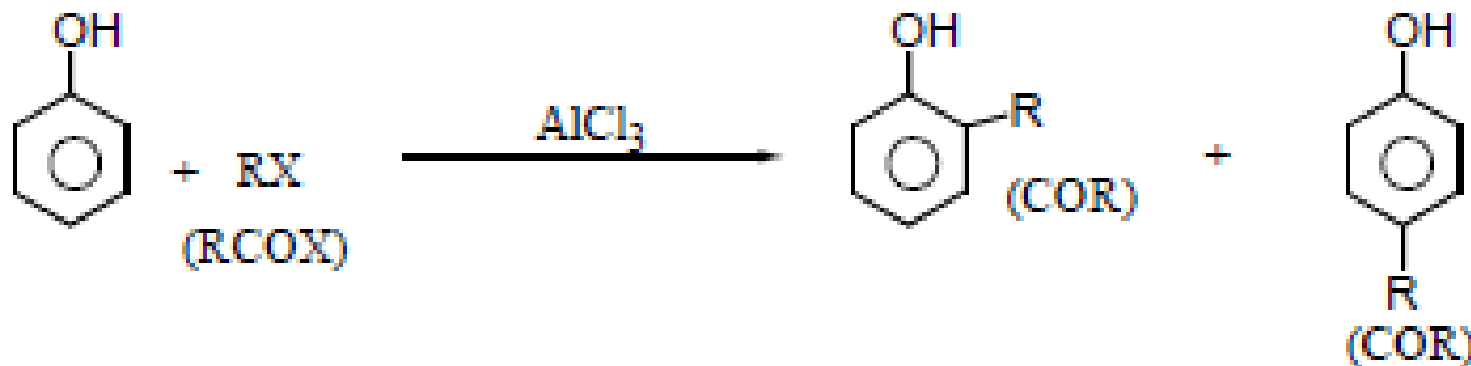
`The precipitate is 2,4,6-tribromophenol.



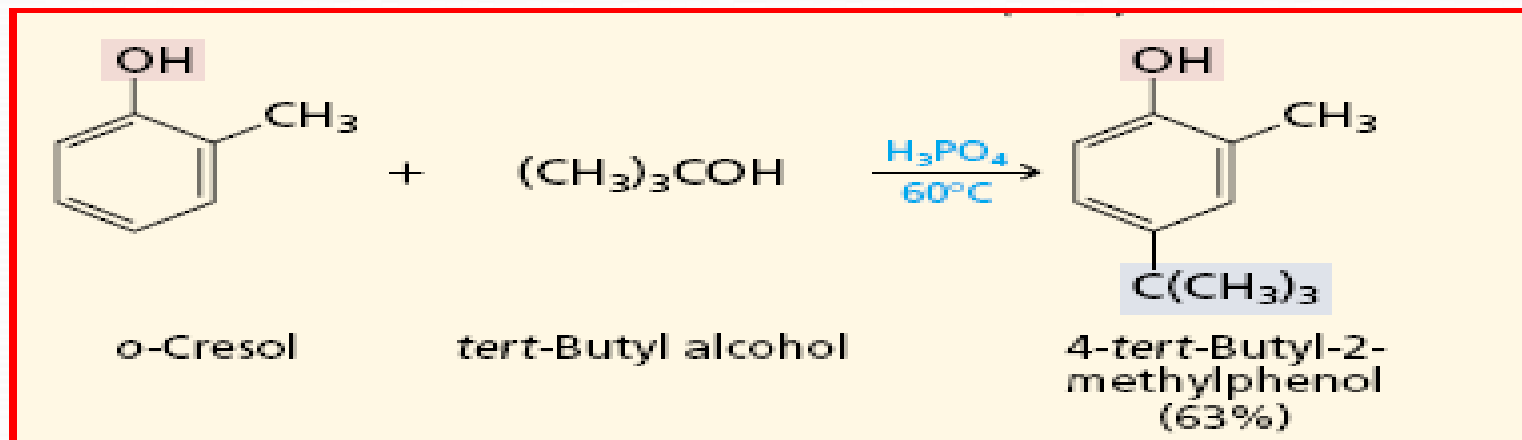
2,4,6-tribromophenol

d- Friedel Crafts Reactions

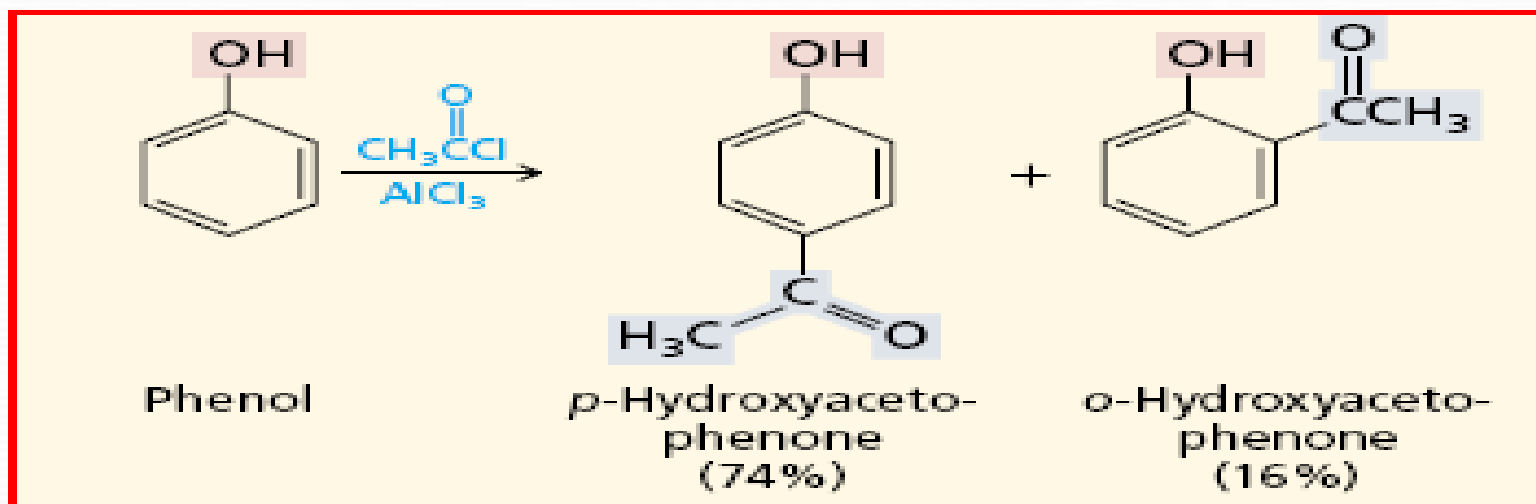
- Reaction of alkyl halide with phenol in the presence of Lewis acid gives alkylation, while reaction with alkanoyl chloride gives a ketone. The products obtained are both the o- and p- products.



Friedel Crafts Alkylation



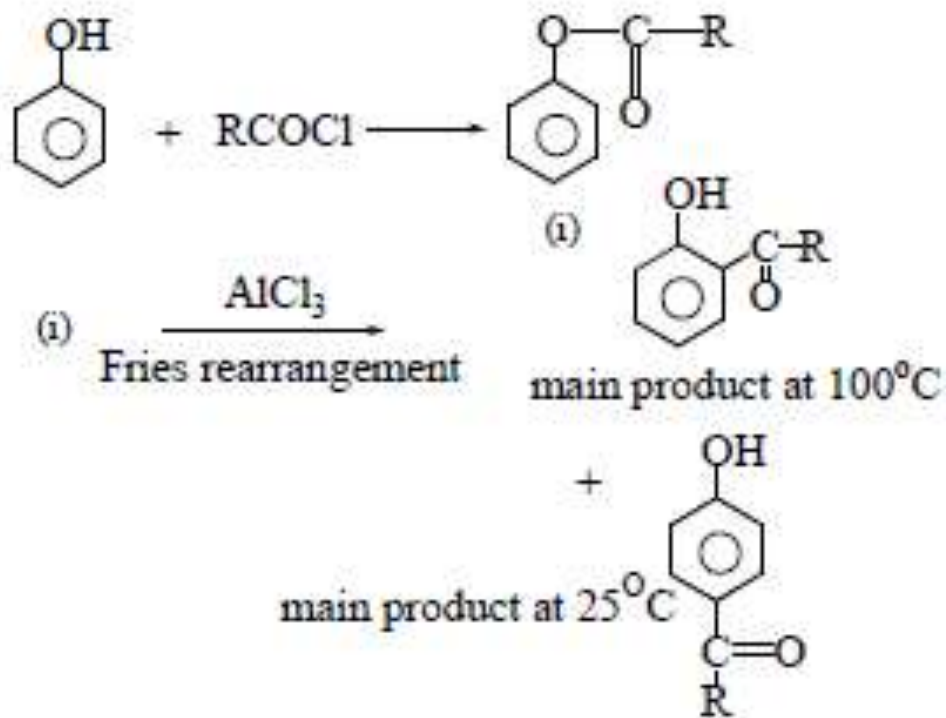
Friedel Crafts Acylation



Friedel Crafts Acylation

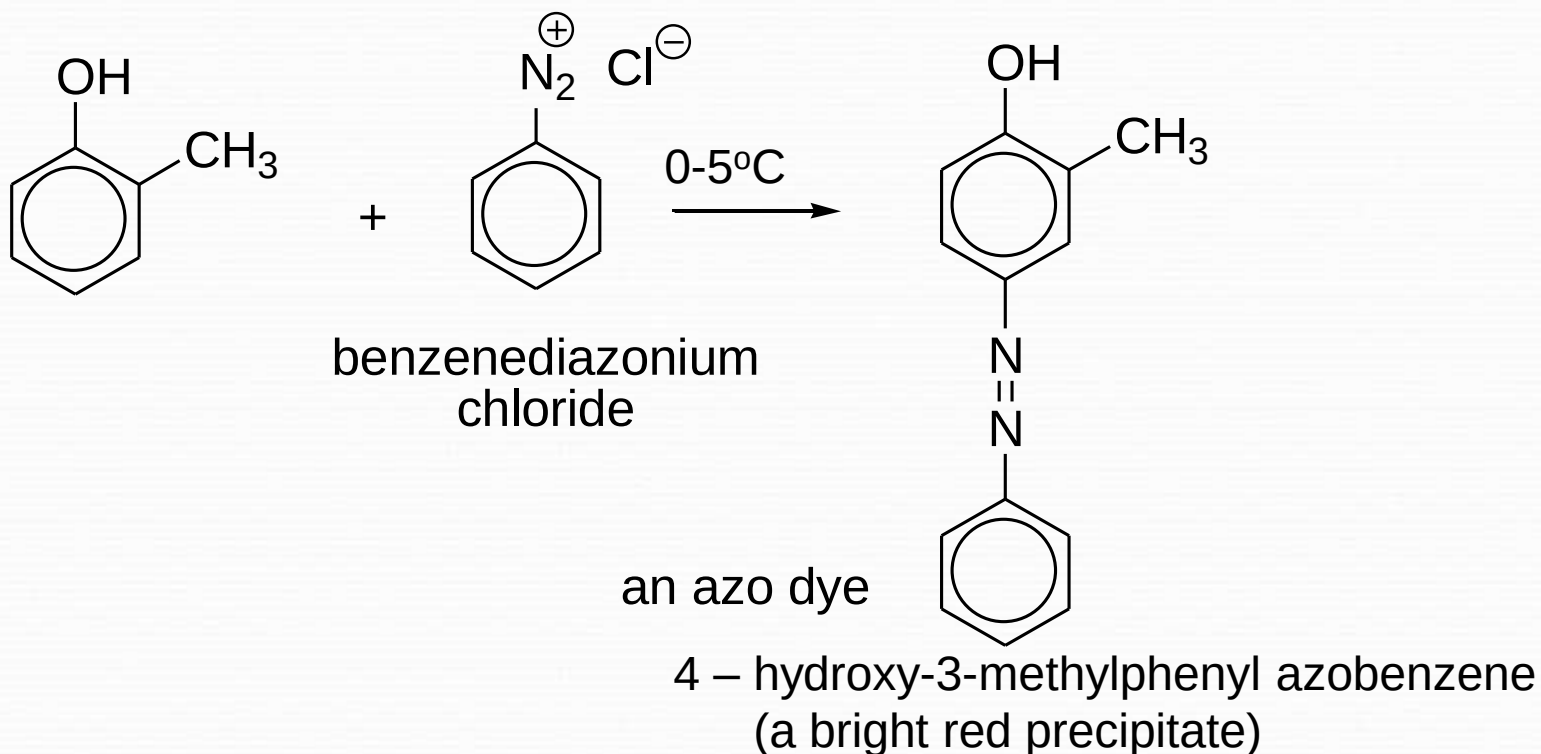
Phenolic esters are often prepared in two steps by means of Fries rearrangement instead of direct acylation. The steps are

- (i) formation of phenyl esters
- (ii) migration of acyl group from phenolic oxygen to ortho or para positions.

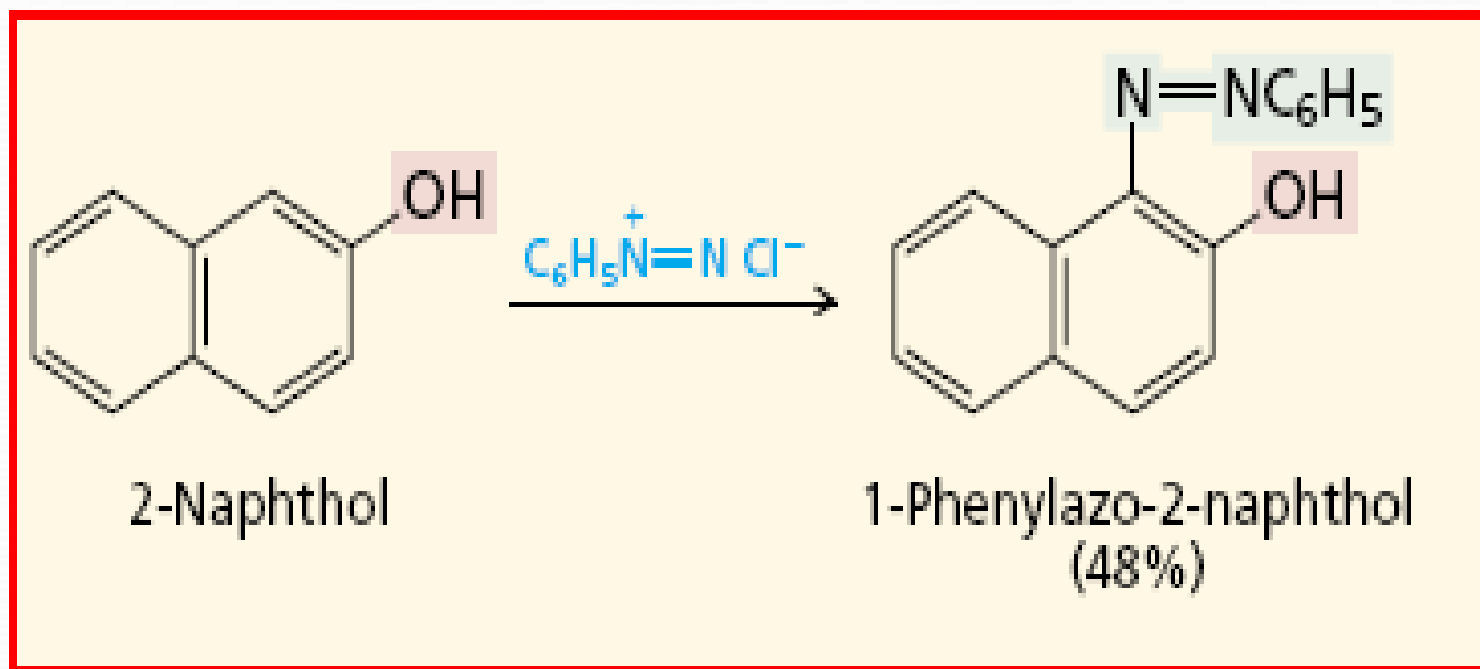


e- Phenols with diazonum salts

Reaction of phenols with diazonum salts at 0-5°C gives azo compounds. This is usually used in identification of aniline (primary aromatic amines).

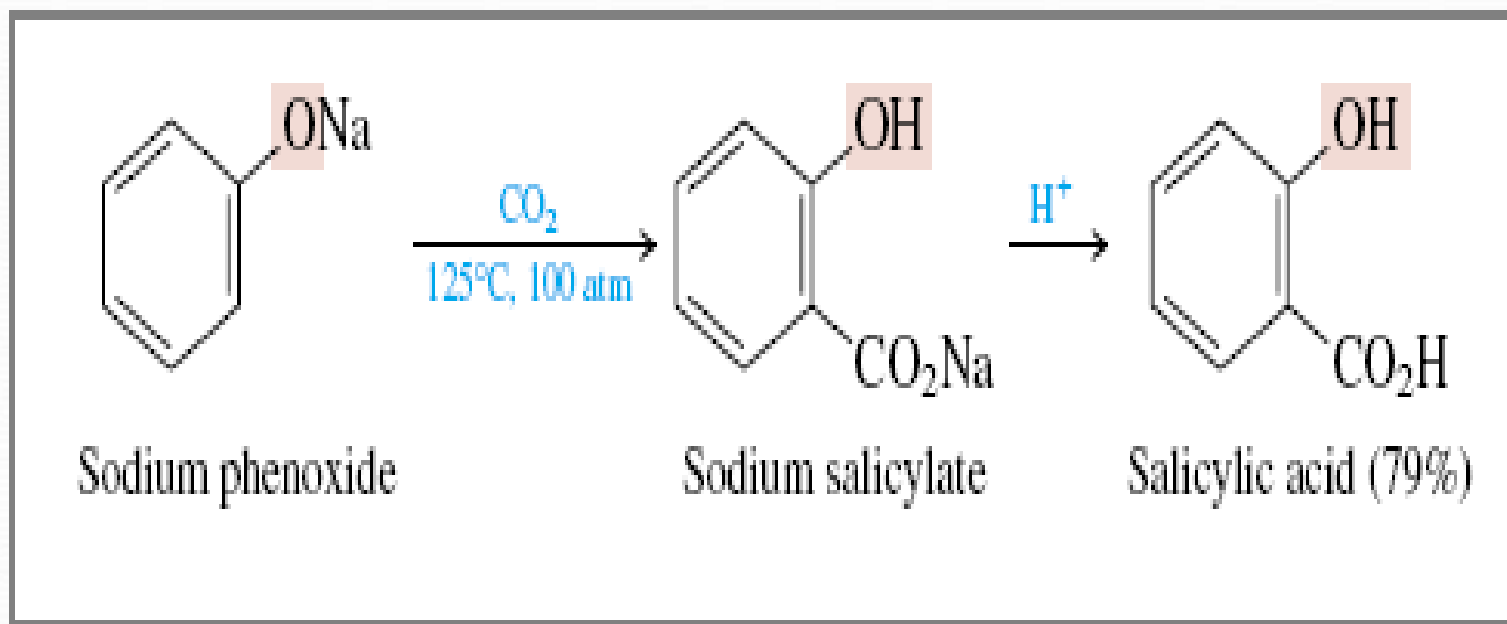


Reaction with arenediazonium salts.



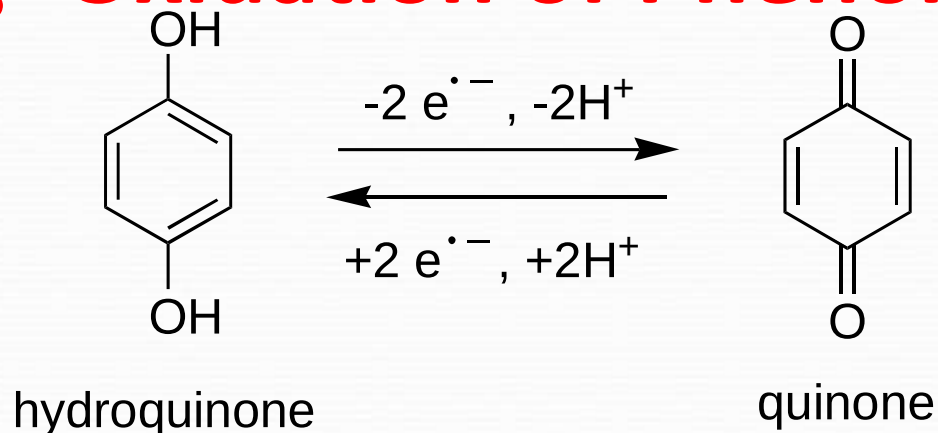
f- Carbonation: Preparation of Salicylic Acid

- Substitution here occurs almost entirely at ortho position.



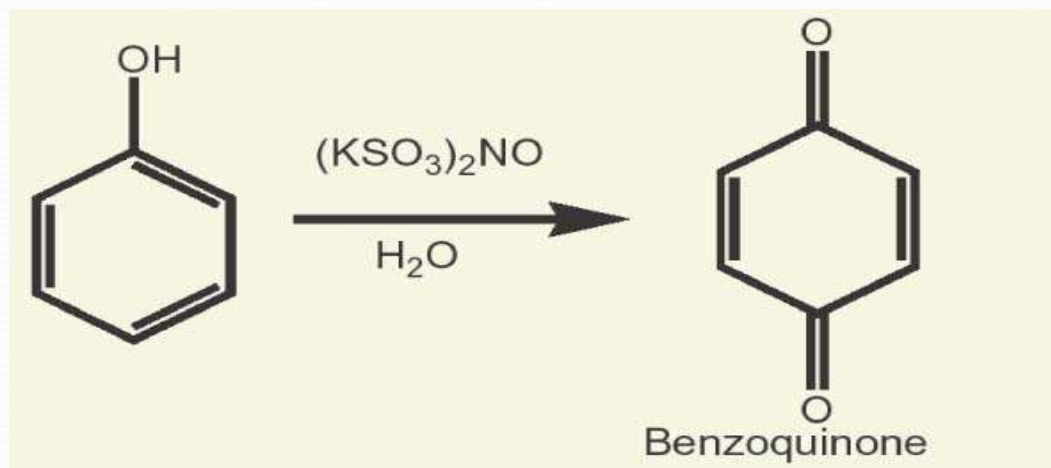
This reaction is called the **Kolbe- Schmitt reaction**. Acidification converts the sodium salt shown above to salicylic acid.

g- Oxidation of Phenols: Quinones



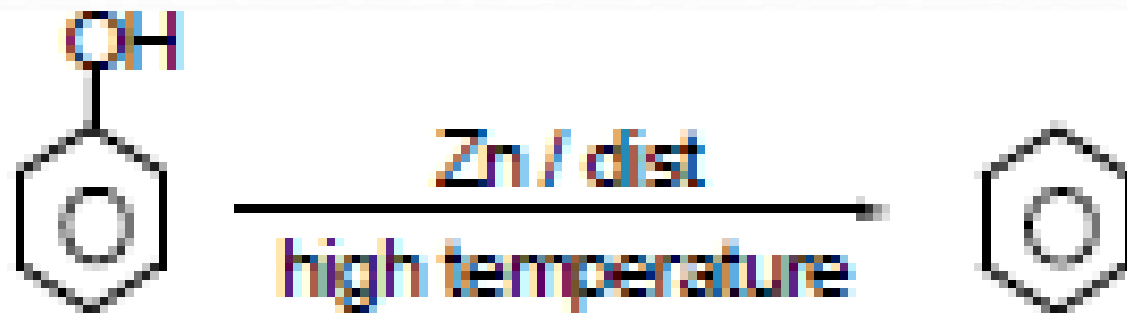
Phenols are easily oxidized to quinones using FeCl_3

Reaction of a phenol with oxidizing agents yields a quinone. Fremy's salt $[(\text{KSO}_3)_2\text{NO}]$ works under mild conditions through a radical mechanism.



h- Reduction

- Phenols are reduced to the corresponding aromatic hydrocarbon by distilling with zinc at high temperature.



i- Nitrosation

